BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DG KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 6487 / 6484

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

OBJECTIVE

Maximum Marks: 17 | **Paper Code:** 6487

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. For a reaction $NaOH + HCl \longrightarrow NaCl + H_2O$ the change in enthalpy is called :

(A) Heat of neutralization
(B) Heat of reaction
(C) Heat of formation
(D) Heat of combustion

Q2. An excess of silver nitrate is added to barium chloride solution and precipitates removed by filtration , what are the main ions in the filtrate ?

(A) Ba^{2+} and NO_3^- only
(B) Ag^+ , Ba^{2+} and NO_3^- only
(C) Ag^+ and NO_3^- only
(D) Ba^{2+} , NO_3^- and Cl^-

Q3. Which of the following solution has the highest boiling point ?

(A) 18 % solution of glucose
(B) 6.0 % solution of urea
(C) 5.85 % solution of sodium chloride
(D) All have the same boiling point

Q4. If a strip of Cu metal is placed in a solution of $FeSO_4$:

(A) Cu will be deposited
(B) Fe is precipitated out
(C) Cu and Fe both dissolved
(D) No reaction takes place

Q5. With increase of 10 °C temperature the rate of reaction doubles. This increase in rate of reaction is due to:

(A) Increase in number of effective collisions.
(B) Increase in activation energy of reactants.
(C) Decrease in activation energy of reaction.
(D) Decrease in the number of collisions between reactant molecules.

Q6. One mole of SO_2 contains :

(A) 6.02×10^{23} atoms of oxygen
(B) 6.02×10^{23} atoms of sulphur
(C) 18.1×10^{23} molecules of SO_2
(D) 4 gram atoms of SO_2

Q7. How many particles are called fundamental particles of an atom ?

(A) 3
(B) 5
(C) 100
(D) 6

Q8. What are the units of R_f value ?

(A) Cm
(B) Cm^3
(C) dm^3
(D) No units

Q9. Which of the following cannot sublime ?

(A) Naphthalene
(B) Iodine
(C) Ammonium chloride
(D) MnO_2

Q10. If absolute temperature of a gas is doubled and the pressure is reduced to one half , the volume of the gas will

(A) Be doubled
(B) Reduced 1/4
(C) Increases four times
(D) Remain unchanged

Q11. Partial pressure of oxygen in lungs (in torr) is :

(A) 150
(B) 116
(C) 760
(D) 159

Q12. Molecules of CO_2 in dry ice form the :

(A) Molecular crystals
(B) Ionic crystals
(C) Covalent crystals
(D) Any type of crystals

Q13. Vapour pressure is not affected by :

(A) Temperature
(B) Intermolecular forces
(C) Surface area
(D) Pressure

Q14. Wave number of the light emitted by a certain source is 2×10^6 m^{-1}. The wavelength of this light will be :

(A) 500 n.m
(B) 500 m
(C) 200 n.m
(D) 5×10^7 m

Q15. Radioactive copper emits :

(A) α - rays
(B) β - rays
(C) γ - rays
(D) Positive rays

Q16. Which of the following molecules have zero dipole moment ?

(A) NH_3
(B) $CHCl_3$
(C) BF_3
(D) H_2O

Q17. The bond order of helium molecule is :

(A) 3
(B) 2
(C) 1
(D) Zero

SUBJECTIVE PART SECTION-I / SECTION-II

Time Allowed: 2 HRS 40 MINUTES | **Maximum Marks:** 68

Q2. (Marks 16) Write short answers to any Eight (8) of the following

i N_2 and CO have the same number of electrons, protons and neutrons, justify.

ii Law of conservation of mass have to be obeyed during stoichiometric calculations, explain.

iii Why actual yield is always less than theoretical yield ?

iv Write two suitable uses of the technique of chromatography

v In solvent extraction technique, why repeated extractions using small portions of solvent are more efficient than using a single extraction but larger volume of solvent.

vi How undesirable colours in crystallization process can be removed ?

vii Write formulas to interconvert various scales of temperature.

viii How density of an ideal gas can be calculated from ideal gas equation ?

ix Derive Charle's law by kinetic equation of gases.

x What is Handerson equation and for what purpose it is used ?

xi What are applications of buffer solutions in daily life ?

xii Derive ionic product of water and what is its value at 25°C .

Q3. (Marks 16) Write short answers to any Eight (8) of the following

i Why intermolecular forces are weaker than intramolecular forces ?

ii What are advantages of Vacuum distillation ?

iii Differentiate between Isomorphism and polymorphism.

iv Diamond is hard and electrical insulator. Justify it.

v Explain Atomic Emission Spectrum.

vi Define (a) Wave number (b) Frequency

vii Write electronic configuration of Cr_{24} and Zn_{30}

viii What is Moseley's law ? Give its mathematical expression.

ix What do you mean by water of crystallization ? Give an example.

x Why NaCl and KNO_3 are used to lower the melting point of ice ?

xi Differentiate between instantaneous and average rate of a reaction.

xii What do you mean by Homogeneous catalysis ? Give an example.

Q4. (Marks 12) Write short answers to any Six (6) of the following

i How does the hybridization scheme explain the bond length ?

ii Define electron affinity. Name the factors affecting it.

iii The radius of an atom cannot be determined precisely. Give the reason.

iv Why do the lone pairs of electrons on an atom occupy more space than bond pairs ?

v Define standard enthalpy of formation. Give an example.

vi Define exothermic reaction. Give an example.

vii Differentiate between spontaneous and non-spontaneous process.

viii What is anodized aluminium ?

ix Give the electrode reactions during the recharging of lead accumulator.

> Note: Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) [1+3] Define limiting reactant , write down the steps involved in identification of limiting reactant.

(B) [1+3] Define hydrogen bonding , how does it explain structure of ice (without diagram).

Q6.

(A) [4] Write a note on " Principal Quantum Number "

(B) [4] 250 Cm^3 of the sample of hydrogen gas effuses four times as rapidly as 250 Cm^3 of an unknown gas. Calculate the molar mass of unknown gas.

Q7.

(A) [1+3] Discuss sp – hybridization with example of ethyne.

(B) [4] The solubility product of Ag_2CrO_4 is 2.6 x 10^{-2} at 25°C. Calculate the solubility of the compound.

Q8.

(A) [4] Describe construction and working of a Bomb Calorimeter.

(B) [4] What is standard electrode potential ? How can it be measured ?

Q9.

(A) [2+2] What are continuous and discontinuous solubility curves ? Draw these curves to explain the answer.

(B) [2+2] Discuss homogeneous and heterogeneous catalysis in detail with two examples of each.

GROUP – II

OBJECTIVE

Maximum Marks: 17 | **Paper Code:** 6484

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. The order of rate of diffusion of gases NH_3 , SO_2 , Cl_2 and CO_2 is

(A) $NH_3 > SO_2 > Cl_2 > CO_2$
(B) $NH_3 > CO_2 > SO_2 > Cl_2$
(C) $Cl_2 > SO_2 > CO_2 > NH_3$
(D) $NH_3 > CO_2 > Cl_2 > SO_2$

Q2. Partial pressure of oxygen in lungs is :

(A) 760 torr
(B) 320 torr
(C) 159 torr
(D) 116 torr

Q3. Which of the following is a Pseudo solid ?

(A) CaF_2
(B) Glass
(C) NaCl
(D) KCl

Q4. The number of Na^+ ions which surround each Cl^- ion in the NaCl crystal is :

(A) 4
(B) 6
(C) 8
(D) 12

Q5. The e/m value for the positive rays is maximum for :

(A) H_2 (B) H_e

(C) O_2 (D) N_2

Q6. The number of neutrons present in $_{19}K^{39}$ is :

(A) 18 (B) 19

(C) 20 (D) 39

Q7. Which of the following has zero dipole moment ?

(A) NH_3 (B) $CHCl_3$

(C) H_2O (D) CO_2

Q8. In Al_2O_3 , the ratio between the ions is :

(A) 1 : 2 (B) 2 : 1

(C) 2 : 3 (D) 3 : 2

Q9. Calorie is equivalent to :

(A) 0.4184 J (B) 41.84 J

(C) 4.184 J (D) 418.4 J

Q10. The pH of human blood is :

(A) 7.0 (B) 7.35

(C) 4.0 (D) 6.5

Q11. In a mixture of 7 g of N_2 and 8 g of O_2 , the mole fraction of O_2 is

(A) 1 (B) 0.1

(C) 0.5 (D) 0.2

Q12. The cell in which electrical energy is converted into chemical energy is called :

(A) Galvanic cell (B) Electrolytic cell

(C) Fuel cell (D) Deniel cell

Q13. Indicate the enzyme which catalyzes the $C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$:

(A) Diastase (B) Zymase

(C) Urease (D) Invertase

Q14. 18 g of H_2O sample has :

(A) 1 mole of H - atom (B) 0.5 mole of O - atom

(C) 6.22×10^{23} moles of H_2O (D) 6.02×10^{23} molecules of H_2O

Q15. The percentage of nitrogen in ammonia is :

(A) (14/34) x 100
(B) (14/17) x 100
(C) (3/17) x 100
(D) (28/38) x 100

Q16. Which one of the following does not undergo sublimation :

(A) $KMnO_4$
(B) Naphthalene
(C) NH_4Cl
(D) Iodine

Q17. The comparative rates at which the solutes move in paper chromatography depend on :

(A) Size of paper
(B) R_f value of solute
(C) Temperature of the experiment
(D) Size of the chromatographic tank used

SUBJECTIVE PART SECTION-I / SECTION-II

Time Allowed: 2 HRS 40 MINUTES | **Maximum Marks:** 68

Q2. (Marks 16) Write short answers to any Eight (8) of the following

i Process of cation formation is endothermic. Justify.

ii What are homoatomic and heteroatomic molecules ? Give one example of each.

iii Why actual yield is always less than theoratical yield ?

iv How rate of filtration can be increased ?

v What is safe and reliable method for drying the crystals ?

vi Give two characteristics of ideal solvent used for crystallization.

vii Define isotherm. What is the effect of temperature on isotherm ?

viii What is quantitative definition of Charle's law ? Give its mathematical form.

ix Define critical temperature. On which factor does it depends

x Define pH and pOH. Give its mathematical form.

xi Define common ion effect. Give one example.

xii What are acidic and basic buffers. Give one example of each.

Q3. (Marks 16) Write short answers to any Eight (8) of the following

i Define Lattice energy. Give example.

ii Why transition temperature is shown by elements having allotropic forms and by compounds showing polymorphism. Give example.

iii Iodine dissolves readily in Tetrachloromethan. Give reason.

iv Water and ethanol can mix easily and in all proportions. Give reason.

v Prove that $E = h c \bar{\nu}$

vi Complete (or) write balanced equation for two Nuclear reactions. (a) $^{4}_{2}He + ^{9}_{4}Be \longrightarrow ?$ (b) $^{14}_{7}N + ^{1}_{0}n \longrightarrow ?$

vii Why is it necessary to decrease the pressure in the discharge tube to get the cathode rays ?

viii How neutrons are used in the treatment of Cancer ?

ix One molal solution of urea in water is dilute as compared to one molar solution of urea, but the number of particles of the solute is same. Justify.

x Differentiate between ideal and non-ideal solutions.

xi The rate of a chemical reaction is an ever changing parameter under the given conditions. Give reason.

xii What is Pseudo first order reaction ?

Q4. (Marks 12) Write short answers to any Six (6) of the following

i Dipole moment of CO_2 is zero , but that of SO_2 is 1.61 D why ?

ii Anionic radius is more than its parent atom why ?

iii Draw geometry of $BeCl_2$ molecule on the basis of VSEPR theory.

iv Define covalent radius. Give one example.

v Define thermochemistry.

vi State standard enthalpy of solution. Give example.

vii Define internal energy.

viii Draw diagram of voltaic cell.

ix Define electrochemistry.

> Note: Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) [1+2+1] What is stoichiometry ? Give its assumptions. Mention two laws which help to perform the stoichiometric calculation

(B) [1+3] Define vapour pressure of liquids. Also explain manometric method for its determination.

Q6.

(A) [4] Calculate the density of CH_4 (g) at 0 °C and 1 atmospheric pressure.

(B) [4] Describe Millikan's oil drop method to measure the charge on electron.

Q7.

(A) [4] . Write down the four postulates of VSEPR theory.

(B) [4] N_2 (g) and H_2 (g) combine to give NH_3 (g). The value of K_c in this reaction at 500 °C is 6.0 x 10^{-2}. Calculate the value of K_p for this reaction.

Q8.

(A) [4] Define the following with examples. (i) Enthalphy (ii) Exothermic reaction (iii) Boundary (iv) Enthalpy of atomization

(B) [4] Write any four industrial importance of electrolytic process.

Q9.

(A) [4] Derive a relationship for $M_2 = (K_b / \Delta T_b) . (1000W_2 / W_1)$

(B) [1+3] What do you the mean by the term " order of reaction " ? Explain by giving any three suitable examples.